

Multi-Modal Fusion of MRI and Genomic Data for Brain Tumor Subtyping

(2024)

Problem

Classifying glioma molecular subtypes to guide treatment planning.

Method

Fusion of 3D CNN features extracted from MRI scans with genomic expression data via a joint embedding layer.

Dataset

TCGA-GBM/LGG genomic cohort combined with the BraTS MRI dataset.

Evaluation

Balanced accuracy and survival correlation analysis.

Results

The fused model achieved 84% balanced accuracy versus 76% for an MRI-only baseline.

Limitations

Small genomic cohort size limits statistical power for rare molecular subtypes.